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CHAPTER 1 • NUMBER • TOPIC 1

Number Toolkit

The number words every question uses —
factors, primes, place value, negatives —
and how to decide a money problem.



$$\frac{3}{10} = 0.3 =$$

30%

5

VARIANTS

ALL

EVERY PAPER USES THESE

38

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Factors & multiples

V2 Primes, squares, cubes

V3 Place value

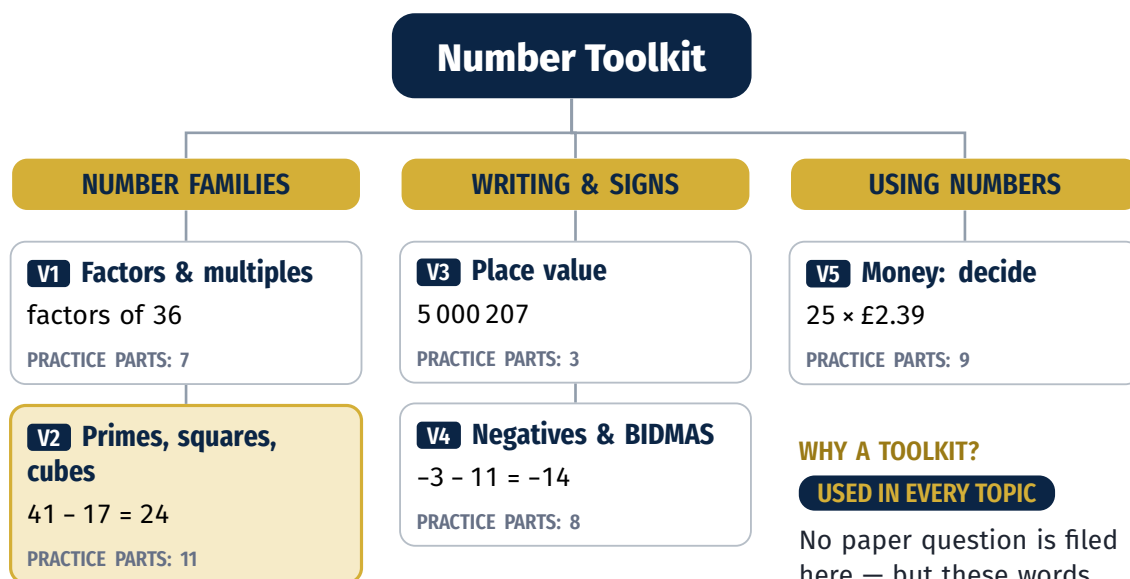
V4 Negatives & BIDMAS

V5 Money: decide

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Number Toolkit at a glance



Your notes


★ How often does each variant come up?

No past-paper question is filed under this topic. 38 practice parts, grouped by type.

	PRACTICE PARTS
V1 Factors & multiples	7
V2 Primes, squares, cubes	11
V3 Place value	3
V4 Negatives & BIDMAS	8
V5 Money: decide	9

WHAT THIS TELLS YOU These are the 1-mark warm-ups and the words inside longer questions (“a prime number between 50 and 60”). Learn the families by heart.

BEFORE YOU START

The number words



Your notes
Integer

..., -2, -1, 0, 1, 2, ...

A whole number — positive, negative or 0.

Factor

12 : 1, 2, 3, 4, 6, 12

Divides **into** the number exactly.**Multiple**

7 : 7, 14, 21, 28, ...

In the **times table** of the number.**Prime**

2, 3, 5, 7, 11, 13, ...

Exactly **two** factors: 1 and itself. 1 is not prime.**Square**

1, 4, 9, 16, 25, 36, ...

 $n \times n$.**Cube**

1, 8, 27, 64, 125, ...

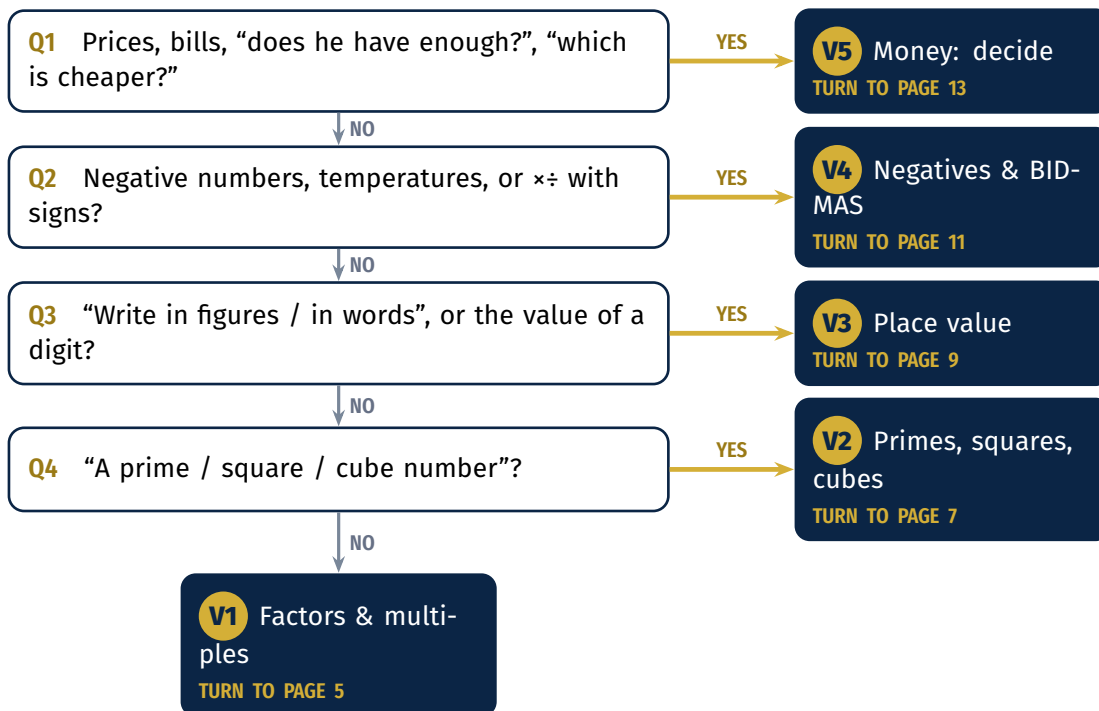
 $n \times n \times n$.**BIDMAS — the order of operations****B** Brackets**I** Indices**D** Divide**M** Multiply**A** Add**S** Subtract $3 \times (-5) + 20 = -15 + 20 = 5$: multiply first, then add.**Sign rules** $+$ $\times$ $+$ $= +$ $3 \times 4 = 12$ $-$ $\times$ $-$ $= +$ $-12 \div -2 = 6$ $+$ $\times$ $-$ $= -$ $3 \times (-5) = -15$ $-$ $\times$ $+$ $= -$ $-8 \div 2 = -4$ **CALCULATOR FIRST, THEN CHECK**Use the **(-)** key for a negative number and brackets for a negative inside a calculation:**((-) 5) .** **SHIFT** **FACT** gives the prime factors of a number — a quick prime test.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Read the question, then walk down the ladder. The first **YES** tells you the variant – and the page to turn to.

**COLOUR CODE IN THE WORKING**

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

What is it?



How do I do it?



Worked example



Check yourself



When you see → you do



Watch out



Exam tip



Calculator check

V1

VARIANT 1

Factors & Multiples

Factors divide into a number; multiples are in its times table.

PRACTICE TOPIC
NO PAST-PAPER QUESTION OF ITS OWN
USED INSIDE OTHER QUESTIONS
PRACTICE QUESTIONS: 7

 Your notes

? What is it?

In plain words

Factors of 36 are the numbers that divide 36 exactly. Find them in **pairs**:

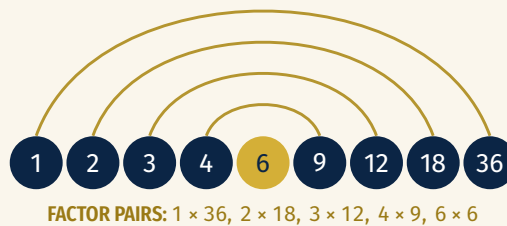
1×36 , 2×18 , ...

Multiples of 7 are 7, 14, 21, 28, ... — the 7 times table, going on for ever.

IT LOOKS LIKE

factors of 36

a multiple of 8



How do I do it?

1 Start at 1

1×36 . Write both numbers of the pair.

2 Go up

Try 2, 3, 4, ... — does it divide exactly? Write the pair.

3 Stop

When the pairs meet (6×6). You have them all.

4 Multiples

For multiples, count up in the times table.

FACTORS COME IN PAIRS A square number has one pair that repeats.

V1 Factors & Multiples

VARIANT 1 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Write down all the factors of 36” → **Factor pairs from 1 until they meet**

“From the list, write down a multiple of 8” → **The number in the 8 times table**

“Why is 0 an even number?” → **$0 = 2 \times 0$: a multiple of 2**

WORKED EXAMPLE **PRACTICE**

Number Toolkit • Easy Q15 • 2 marks

Complete the list of factors of 36: 1, 2, ..., 36

- | | | |
|---------------|----------------------------------|------------------------|
| STEP 1 | $1 \times 36, \quad 2 \times 18$ | ◀ start at 1, then 2 |
| STEP 2 | $3 \times 12, \quad 4 \times 9$ | ◀ 5 does not divide 36 |
| STEP 3 | 6×6 | ◀ the pairs meet: stop |

ANSWER 1, 2, 3, 4, 6, 9, 12, 18, 36

✓ 9 factors

WATCH OUT

✗ factors of 36: 1, 2, 3, 4, 6, 36

✓ also 9, 12, 18

Write the **pair** each time — the big partner is easy to forget.

EXAM TIP

“Complete the list” already shows 1, 2 and 36 — your answer is the missing ones in order.

Check yourself

calculator first • answers on page 15

1.1 Practice • Easy

From 32, 33, ..., 39 write down a multiple of 8.

Answer:

1.2 Practice • Easy

Give one reason why 0 is an even number.

Answer:

1.3 Dr Eslam

Write down all the factors of 24.

Answer:



VARIANT 2

Primes, Squares & Cubes

Pick the special numbers out of a list.

PRACTICE TOPIC
NO PAST-PAPER QUESTION OF ITS OWN
USED INSIDE OTHER QUESTIONS
PRACTICE QUESTIONS: 11



Your notes

? What is it?

In plain words

Prime: exactly two factors (1 and itself). 1 is **not** prime; 2 is the only even prime.

Square:

1, 4, 9, 16, 25, 36, 49, 64, 81, 100.

Cube: 1, 8, 27, 64, 125.

IT LOOKS LIKE

a prime between 50 and 60

a cube between 50 and 100

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

PRIME

SQUARE

How do I do it?

1 Know the lists

Primes to 50, squares to 100, cubes to 125.

2 Test a prime

Try dividing by 2, 3, 5, 7. None works? It is prime.

3 Read the range

“Between 50 and 60” — check every number in it.

4 Answer what is asked

One number, or the difference between two.

1 IS NOT PRIME It has only one factor.

V2 Primes, Squares & Cubes

VARIANT 2 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Write down a prime number between 50 and 60" → **Test 51–59: 53 or 59**

"A cube number greater than 50 but less than 100" → $4^3 = 64$

"The difference between the two primes in the list" → **Find both, subtract**

WORKED EXAMPLE **PRACTICE**

Number Toolkit • Medium Q11 • 2 marks

From the list 22, 17, 25, 41, 39, 4 work out the difference between the two prime numbers.

STEP 1 $22 = 2 \times 11$, $25 = 5^2$, $39 = 3 \times 13$, $4 = 2^2$ ◀ not prime

STEP 2 17, 41 ◀ only factors 1 and itself

STEP 3 $41 - 17 = 24$ ◀ the difference

ANSWER 24

✓ 17 and 41 are prime

WATCH OUT

✗ 39 is prime

✓ $39 = 3 \times 13$

Test small primes: 3 divides 39. Odd does not mean prime.

EXAM TIP

SHIFT **FACT** on the calculator breaks a number into primes — a prime shows just itself.

Check yourself

calculator first • answers on page 15

2.1 Practice • Medium

Write down the cube number greater than 50 but less than 100.

Answer:

2.2 Practice • Medium

Write down a prime number between 50 and 60.

Answer:

2.3 Practice • Easy

From 87, 77, 57, 47, 27 write down a cube number and a prime number.

Answer:

V3

VARIANT 3

VARIANT 3

Place Value

Each digit's value depends on its column.

PRACTICE TOPIC

**NO PAST-PAPER QUESTION OF ITS OWN
USED INSIDE OTHER QUESTIONS**

PRACTICE QUESTIONS: 3



Your notes

? What is it?

In plain words

Read big numbers in

groups of three from the right: millions, thousands, units. A missing column is a **zero**.

“Five million, two hundred and seven”: 5 in the millions, 207 in the units, zeros between.

M	HTh	TTh	Th	H	T	U
5	0	0	0	2	0	7
MILLIONS			THOUSANDS		UNITS	

IT LOOKS LIKE

5 000 207 15 060

How do I do it?

- 1 Draw the columns** M · HTh · TTh · Th · H · T · U.

- 2 Fill each group** Put each part of the words in its group.

- ### 3 Zeros
- Every empty column gets a 0.

- 4 Check** Read it back aloud.

GROUPS OF THREE 5 000 207 — a space every three digits.

V3 Place Value

VARIANT 3 • SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO

“Write ... in figures” → **Columns, then zeros in every gap**

“Write 15 060 in words” → **Read the thousands group, then the rest**

“The value of the 7 in 4 728 103” → **Its column: 700 000**

WORKED EXAMPLE **PRACTICE**

Number Toolkit • Easy Q13 • 1 mark

Write the number five million, two hundred and seven in figures.

STEP 1 columns M • HTh • TTh • Th • H • T • U ◀ draw the columns

STEP 2 millions: 5 units group: 207 ◀ fill each group

STEP 3 thousands group empty: 000 ◀ zeros

ANSWER 5 000 207

✓ read back: five million, two hundred and seven

WATCH OUT

✗ 5 207 or 500 207

✓ 5 000 207

Each group has three digits — the empty thousands group is 000.

EXAM TIP

Count the digits: a number in the millions has 7 digits.

Check yourself

calculator first • answers on page 15

3.1 Practice • Easy

Write two hundred thousand and seventeen in figures.

Answer:

3.2 Practice • Easy

Write 15 060 in words.

Answer:

3.3 Dr Eslam

What is the value of the 7 in 4 728 103?

Answer:



VARIANT 4

Variant 4

Temperatures and signs: use a number line and the sign rules.

PRACTICE TOPIC
NO PAST-PAPER QUESTION OF ITS OWN
USED INSIDE OTHER QUESTIONS
PRACTICE QUESTIONS: 8



Your notes

? What is it?

In plain words

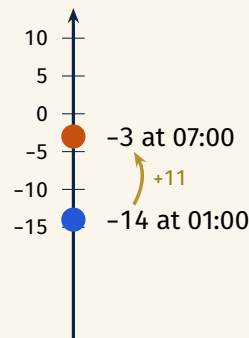
A **number line** (or a thermometer) shows it: **up** is adding, **down** is subtracting.

× and ÷: **same signs → positive**,
different signs → negative. And
BIDMAS decides the order.

IT LOOKS LIKE

-3 - 11

21 - 26

$$-12 \div -2$$


How do I do it?

1 Picture it

Draw a number line; mark the start.

2 Direction

Rise / higher / add → up. Fall / lower / subtract → down.

3 Signs

$\times \div$: same signs +, different signs -.

4 Order

BIDMAS: brackets, powers, $\times \div$, then $+-$.

SAME SIGNS +, DIFFERENT - For $\times$ and $\div$ only.

V4 Negatives & BIDMAS

VARIANT 4 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"... is 11°C higher than the temperature at 01:00" → **Go down 11 to find 01:00**

"had fallen by 26°C " → **Subtract:** $21 - 26$

"Calculate $-12 \div -2$ " → **Same signs:** $+6$

WORKED EXAMPLE **PRACTICE**

Number Toolkit • Easy Q12 • 1 mark

The temperature at 07:00 is -3°C . This is 11°C higher than the temperature at 01:00. Find the temperature at 01:00.

STEP 1 start at -3 (07:00) ◀ picture it

STEP 2 07:00 is higher, so 01:00 is 11 **lower** ◀ direction

STEP 3 $-3 - 11 = -14$ ◀ subtract

ANSWER -14°C ✓ $-14 + 11 = -3$ ✓**WATCH OUT**✗ $-3 + 11 = 8$ ✓ $-3 - 11 = -14$

The later temperature is the higher one
— go back down to find the earlier one.

EXAM TIP

Check by doing it the other way: from your answer, go up 11 — do you reach -3 ?

Check yourself

calculator first • answers on page 15

4.1 Practice • Easy

Noon was 21°C . By midnight it had fallen by 26°C . Find the midnight temperature.

Answer:

4.2 Practice • MediumCalculate $-12 \div -2$

Answer:

4.3 Practice • Medium

The lowest is -57.0°C ; the highest is 63.8°C more. Find the highest.

Answer:

V5

VARIANT 5

**Money Problems:
Decide**

Work out both totals, compare, and say clearly which.

PRACTICE TOPIC
NO PAST-PAPER QUESTION OF ITS OWN
USED INSIDE OTHER QUESTIONS
PRACTICE QUESTIONS: 9
Your notes**? What is it?****In plain words**

A choice: “Is it enough?”, “Which is cheaper?”. Work out

each option in full (same units — all £ or all p), then **compare** and **say** the decision with the numbers.**IT LOOKS LIKE** $£2.39 \times 25$ $£52.50 + 20\% \text{ VAT}$ Kirsty's
£2.39 each
 $\times 25$
£59.75Hedge World
£52.50
 $+20\% \text{ VAT}$
£63.00**CHEAPER ✓****How do I do it?****1 Same units**

All in £ or all in pence.

2 Option A

Work it out fully (multiply, add VAT, ...).

3 Option B

Work it out fully.

4 Decide

Compare and write a sentence with both numbers.

NUMBERS + A SENTENCE “Yes/No” alone scores nothing.

V5 Money Problems: Decide

VARIANT 5 • SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO*"Does he have enough money?"*→ **Total cost vs money; say yes/no with both***"How much cheaper?"*→ **Both totals, then subtract***"plus VAT at 20%"*→ **×1.2****WORKED EXAMPLE** **PRACTICE**

Number Toolkit • Medium Q4 • 5 marks

Tom buys 25 plants. Kirsty's Plants: £2.39 each. Hedge World: pack of 25 for £52.50 plus VAT at 20%. Which should he choose?

STEP 2 $25 \times 2.39 = 59.75$

◀ Kirsty's total

STEP 3 $52.50 \times 1.2 = 63.00$

◀ Hedge World with VAT

STEP 4 $£59.75 < £63.00$, so Kirsty's is cheaper

◀ decide in a sentence

ANSWER Kirsty's Plants (£59.75 vs £63.00)

✓ both totals shown

WATCH OUT

✗ forgetting the VAT

✓ $52.50 \times 1.2 = 63.00$

"Plus VAT" must be added before you compare.

EXAM TIP

Write money with two decimal places:
£9.00, not £9.

Check yourself

calculator first • answers on page 15

5.1 Practice • Easy

Readings 7155 and 7095 units, 15p per unit. How much does Gary pay, in pounds?

Answer:

5.2 Practice • Medium

27 plants at £5.54 each. Steve has £150. Is that enough?

Answer:

5.3 Practice • Medium

18 day returns at £6.45 or a monthly saver at £98.50. How much cheaper is the saver?

Answer:

CHECK YOURSELF

Answers



Every answer below was checked with sympy (check_answers.py). Got one wrong? Go back to that variant's Watch-out box first.

1.1 32**1.2** $0 = 2 \times 0$: a multiple of 2**1.3** 1, 2, 3, 4, 6, 8, 12, 24**2.1** 64**2.2** 53 or 59**2.3** cube 27, prime 47**3.1** 200 017**3.2** fifteen thousand and sixty**3.3** 700 000**4.1** -5°C **4.2** 6**4.3** 6.8°C **5.1** £9.00**5.2** yes: £149.58, 42p left**5.3** £17.60 cheaper

THE WHOLE TOPIC ON ONE PAGE

Number Toolkit cheat sheet



Your notes
V1 Factors & multiples

36 : 1, 2, 3, 4, 6, 9, 12, 18, 36

Factor pairs from 1 until they meet.

V2 Primes, squares, cubes

2, 3, 5, 7, 11 · 4, 9, 16 · 8, 27, 64

1 is not prime; 2 is the only even prime.

V3 Place value

5 000 207

Groups of three; zeros in every gap.

V4 Negatives & BIDMAS $-3 - 11 = -14$

Number line; same signs +; BIDMAS order.

V5 Money: decide

£59.75 < £63.00

Both totals, same units, a sentence.

**TOP 3 LOST MARKS**

1. calling 1 (or 39) prime ×
2. losing a factor pair ×
3. "Yes" with no numbers ×

BOTTOM LINE Know the families · columns for place value · a number line for negatives · both totals before you decide.